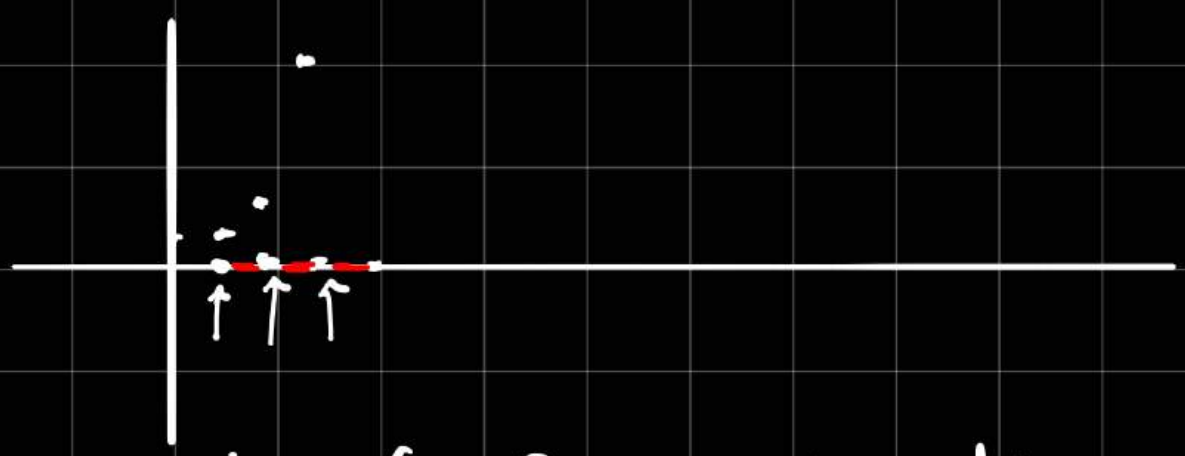
Topics to Cover:

1. Gamma Function.
2. Gaussian Integral.
3. Stirling's Appponimation.
4. Lagrange Multiplier.
5. Counting and Combinatories.
6. Fourier Series and transformation.

Gamma Function:

The Def^n of Gamma function is

$$\Gamma(n+1) = \int_0^\infty e^{-x} x^n \, dx$$



Motivation for Gamma Function:

Factorial isn't defined for the intervals marked red.

So what we need is a more generalized version of the factorial function that is defined for all real numbers. This is where the gamma function comes in.

Solution of the Gamma Function:

$$\Gamma(n+1) = \int_0^\infty e^{-x} x^n \, dx$$

By performing integration by parts we get

$$-e^{-x} x^n \Big|_0^\infty + \int_0^\infty n x^{n-1} e^{-x} \, dx$$

$$= -\left[\lim_{x\to\infty}\left(\frac{x^n}{e^x}\right) - \frac{0^n}{e^0}\right] + \int_0^\infty n x^{n-1} e^{-x} \, dx$$

$$= -\left[\lim_{n\to\infty} \frac{n!}{e^x} - 0\right] + n \int_0^\infty e^{-x} x^{n-1} \, dx$$

0

$$\Rightarrow \int_0^\infty e^{-x} x^n dx = n \int_0^\infty e^{-x} x^{n-1} dx$$

$$\boxed{\therefore \Gamma(n+1) = n\Gamma(n)}$$ → The Recursion Relation.

By performing the same integration by parts 'n' times, we get the following

$$\int_0^\infty e^{-x} x^n dx = n! \int_0^\infty e^{-x} dx$$

$$= n! \left(-e^{-x}\right)\Big|_0^\infty$$

$$= -n! \left(\lim_{x \to \infty} e^{-x} - e^0\right)$$

$$= -n! \left(0 - 1\right)$$

$$= n!$$

$\therefore \Gamma(n+1) = n!$ (Proven)

The Gamma function is also called the generalized factorial function.

Now $\Gamma\left(\frac{3}{2}\right) = \Gamma\left(\frac{1}{2}+1\right) = \left(\frac{1}{2}\right)!$

$$= \int_0^\infty e^{-x} x^{\frac{1}{2}} \, dx$$

$x^{\frac{1}{2}} = u$

$x = u^2$

$dx = 2u\,du$

$dx = 2x^{\frac{1}{2}} du$

$$= \int_0^\infty e^{-u^2} u \, (2u\,du)$$

$$= 2\int_0^\infty u^2 e^{-u^2} du$$

$$= \frac{1}{2}\sqrt{\pi}$$

$$= \frac{\sqrt{\pi}}{2}$$

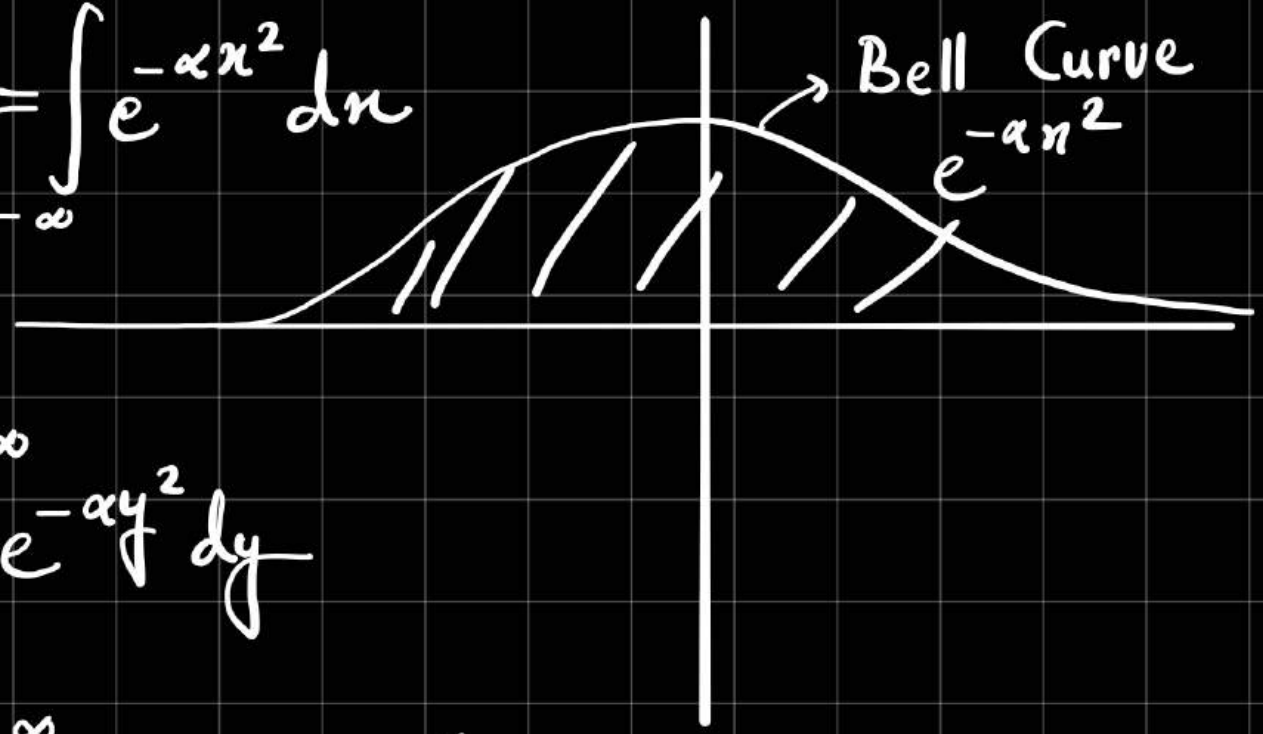
$$\left(\frac{1}{2}\right)! = \frac{\sqrt{\pi}}{2}$$

$$\left(\frac{3}{2}\right)! = \frac{3}{2}\cdot\left(\frac{1}{2}\right)! = \frac{3\sqrt{\pi}}{4}$$

Since we already invoked the Gaussian integral, we might as well delve into that.

② Gaussian integral:

Let $I = \int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx$



$$I = \int_{-\infty}^{\infty} e^{-\alpha y^2}\,dy$$

$$I^2 = \int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx \int_{-\infty}^{\infty} e^{-\alpha y^2}\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\alpha(x^2+y^2)}\,dx\,dy$$

$$\boxed{\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \\ dx\,dy &= r\,dr\,d\theta \end{aligned}}$$

$$= \int_{0}^{2\pi}\int_{0}^{\infty} e^{-\alpha r^2}\, r\,dr\,d\theta$$

$$\boxed{\begin{aligned} \alpha r^2 &= u \\ 2\alpha r\,dr &= du \\ r\,dr &= \frac{du}{2\alpha} \end{aligned}}$$

$$= \int_{0}^{2\pi}\int_{0}^{\infty} e^{-u}\,\frac{du}{2\alpha}\,d\theta$$

$$= \frac{1}{2\alpha}(2\pi)\left(-e^{-u}\right)\Big|_{0}^{\infty}$$

$$= -\frac{\pi}{\alpha}(0-1)$$

$$= \frac{\pi}{\alpha}$$

$$I^2 = \frac{\pi}{\alpha}$$

$$\therefore I = \sqrt{\frac{\pi}{\alpha}}$$

$$\int_{-\infty}^{\infty} e^{-\alpha x^2}\,dx = \sqrt{\frac{\pi}{\alpha}}$$

But what about $\int_0^\infty x^{2n+1} e^{-\alpha x^2} dx$ and $\int_0^\infty x^{2n} e^{-\alpha x^2} dx$?

We can use Leibniz's rule of integration for that. This is where we parameterize the integral as function of another introduced parameter, and differentiate it with respect to that parameter

So now $I(\alpha) = \int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \sqrt{\frac{\pi}{\alpha}}$

$$\Rightarrow \frac{d}{d\alpha} I(\alpha) = \frac{d}{d\alpha} \int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \frac{d}{d\alpha} \sqrt{\frac{\pi}{\alpha}}$$

$$\therefore I'(\alpha) = \int_{-\infty}^{\infty} x^2 e^{-\alpha x^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{\alpha^3}} \rightarrow 1^{st} \text{ Derivative}$$

Similarly

$$I''(\alpha) = \int_{-\infty}^{\infty} x^4 e^{-\alpha x^2} dx = \left(\frac{1}{2}\right)\left(\frac{3}{2}\right) \sqrt{\frac{\pi}{\alpha^5}} \rightarrow 2^{nd}$$

$$\int_{-\infty}^{\infty} x^6 e^{-\alpha x^2} dx = \left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{5}{2}\right) \sqrt{\frac{\pi}{\alpha^7}} \rightarrow 3^{rd}$$

$$\int_{-\infty}^{\infty} x^8 e^{-\alpha x^2}\,dx = \left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(\frac{5}{2}\right)\left(\frac{7}{2}\right)\sqrt{\frac{\pi}{\alpha^9}} \rightarrow 4^{th}$$

$\vdots$ Doing it n times we get

$$\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2}\,dx = \frac{1\cdot 3\cdot 5\cdot 7\cdot \cdots (2n-1)}{2^n}\sqrt{\frac{\pi}{\alpha^{2n+1}}}$$

$$= \frac{1\cdot 2\cdot 3\cdot 4\cdot 5\cdot 6\cdot 7\cdot 8\cdot \cdots (2n-1)\cdot 2n}{2^n\; 2\cdot 4\cdot 6\cdot 8\cdot \cdots 2n}\sqrt{\frac{\pi}{\alpha^{2n+1}}}$$

$$= \frac{(2n)!\;\sqrt{\frac{\pi}{\alpha^{2n+1}}}}{2^n(2\cdot 1)(2\cdot 2)\cdot(2\cdot 3)(2\cdot 4)\cdots(2\cdot n)}$$

$$= \frac{(2n)!\;\sqrt{\frac{\pi}{\alpha^{2n+1}}}}{2^n(2\cdot 2\cdot 2\cdot 2\cdots 2)(1\cdot 2\cdot 3\cdots n)}$$

$$\boxed{\int_{-\infty}^{\infty} x^{2n} e^{-\alpha x^2}\,dx = \frac{(2n)!}{4^n\, n!}\sqrt{\frac{\pi}{\alpha^{2n+1}}}}$$

With that out of the way we can do the following $\int_0^{\infty} x e^{-\alpha x^2}\,dx$

$$= \int_0^\infty \frac{du}{2\alpha} e^{-u}$$

$$= \frac{1}{2\alpha}$$

$$\int_0^\infty x e^{-\alpha x^2} dx = \frac{1}{2\alpha}$$

$$\frac{d}{d\alpha} \int_0^\infty x e^{-\alpha x^2} dx = \frac{d}{d\alpha}\left(\frac{1}{2\alpha}\right)$$

$$\Rightarrow \int_0^\infty x^3 e^{-\alpha x^2} dx = \frac{1}{2}\left(\frac{1}{\alpha^2}\right) \rightarrow 1^{st} \text{ Derivative}$$

$$\Rightarrow \int_0^\infty x^5 e^{-\alpha x^2} dx = \frac{1}{2} \; 1 \cdot 2 \; \frac{1}{\alpha^3} \longrightarrow 2^{nd}$$

$$\Rightarrow \int_0^\infty x^7 e^{-\alpha x^2} dx = \frac{1}{2} \cdot 1 \cdot 2 \cdot 3 \; \frac{1}{\alpha^4} \longrightarrow 3^{rd}$$

$$\boxed{\therefore \int_0^\infty x^{2n+1} e^{-\alpha x^2} dx = \frac{n!}{2\alpha^{n+1}}} \; ; n \geq 0$$

The Final boss:

Considering the following integral

$$\int_{-\infty}^{\infty} e^{-ax^2+bx}\,dx \quad \text{for} \quad a>0,\ b\in\mathbb{R}$$

$$= \int_{-\infty}^{\infty} e^{-a\left(x^2-2\cdot\frac{b}{2a}x+\frac{b^2}{4a^2}-\frac{b^2}{4a^2}\right)}\,dx$$

$$= \int_{-\infty}^{\infty} e^{-a\left(x-\frac{b}{2a}\right)^2} e^{\frac{b^2}{4a}}\,dx$$

$$= e^{\frac{b^2}{4a}} \int_{-\infty}^{\infty} e^{-a\left(x-\frac{b}{2a}\right)^2}\,dx$$

$$= e^{\frac{b^2}{4a}} \sqrt{\frac{\pi}{a}}$$

$$= \left(\frac{\pi}{a}\right)^{1/2} e^{\frac{b^2}{4a}}$$

The Generalized formula:

$$\frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}}\,dx = 1$$

Here
μ = mean
σ = Standard Deviation
σ^2 = Variance

③ Stirling's Approximation :

$$N! = N(N-1)(N-2) \cdots\cdots 3 \cdot 2 \cdot 1$$

$$\Rightarrow \ln N! = \ln 1 + \ln 2 + \ln 3 + \cdots\cdots + \ln N$$

$$\therefore \ln N! = \sum_{k=1}^{N} \ln k$$

Now the discrete sum can be converted to an integral if we make the individual distances even smaller. Which is to say

$$\ln N! = \sum_{k=1}^{N} \ln k \, \Delta x_k$$

$$\therefore \lim_{\Delta x_k \to 0} \ln N! \approx \int_1^N \ln x \, dx$$

$$= \left[x \ln x - x \right]_1^N$$

$$= N \ln N - N - 1 \ln 1 + 1$$

$$\therefore \ln N! = N \ln N - N + 1$$

For very large N in other words $N \gg 1$

$$\boxed{\ln N! = N\ln N - N}$$ → Logarithmic Form

What about the exponential form?

Recalling $\ln N! = N\ln N - N$

$$\Rightarrow N! = e^{N\ln N - N}$$

$$\Rightarrow N! = e^{\ln N^N} e^{-N}$$

$$\therefore N! = N^N e^{-N}$$

Now this is the Stirling's formula for large values of N.

We can however add more term for increased accuracy.

We can recall the Gamma function

$$\Gamma(N+1) = \int_0^\infty e^{-x} x^N \, dx = N!$$

Now let $e^{f(x)} = x^n e^{-x}$

$$\therefore f(x) = n\ln x - x$$

$$f'(x) = \frac{n}{x} - 1$$

$$f''(x) = \frac{n}{x^2}$$

Now if we make a Taylor expansion of $f(x)$ centered around n. we get

$$f(x) = f(n) + \left(\frac{df}{dx}\right)\Bigg|_{x=n}(x-n) + \frac{1}{2}\left(\frac{d^2f}{dx^2}\right)\Bigg|_{x=n} + \cdots$$

$$= n\ln n - n + \left(\frac{n}{n} - 1\right)(x-n) + \frac{1}{2}\left(\frac{n}{n^2}\right)(x-n)^2 + \cdots$$

$$= n\ln(n) - n - \frac{(x-n)^2}{2n} + \cdots$$

$$\therefore e^{f(x)} = \exp\left[n\ln(n) - n - \frac{(x-n)^2}{2n} + \cdots\right]$$

$$= e^{n\ln(n) - n}\, e^{-\frac{(x-n)^2}{2n} + \cdots}$$

$$\Rightarrow \int_0^\infty e^{f(x)}dx = \int_0^\infty e^{-x}x^n dx = \int_0^\infty e^{n\ln(n) - n}\, e^{-\frac{(x-n)^2}{2n} + \cdots} dx$$

$$\Rightarrow n! = e^{n\ln(n) - n}\int_0^\infty e^{-\frac{(x-n)^2}{2n}} dx$$

$$\therefore n! = e^{\ln n^n} e^{-n} \sqrt{\frac{\pi}{\frac{1}{2n}}} = n^n e^{-n} \sqrt{2\pi n}$$

This implies

$$\boxed{\begin{aligned} n! &= n^n e^{-n} \sqrt{2n\pi} \\ \ln n! &= n \ln n - n + \frac{1}{2} \ln(2n\pi) \end{aligned}}$$

The Stirling's Approximation.

The Stirling's approximation is used in this scenario because doing it using factorial often remains tedious.

④ Lagrange Multiplier:

Motivation:

Lagrange multipliers are used everywhere in physics to analyze systems that are subject to at least one constraint or more. In Classical Mechanics this often lowers the number of equations needed to solve the problem consisting of 6N co-ordinates, 3N is space and 3N in momentum. Altogether it forms a phase space of 6N co-ordinates on which we can impose these constraints to simplify the solution process.

Now let $x = (x_1, x_2, x_3, \cdots x_n)$ if we have a total of 'k' constraints. we can impose them on the original function f and create a new one F.

Here $F = f + \sum_{k=1}^{\text{total Constraints}} \lambda_k g_k$

Here λ_k = Lagrange Multiplier

and g_k = The function defining the constraint

We can solve for λ_k using $\frac{\partial F}{\partial x_k} = 0$

Enough with Theories. Let's look at an example.

Problem: Find the ratio of the base radius of a solid cone to it's height that maximizes the total surface area subject to a constraint of constant volume.

Sol^n:

We know that for the cone.

Base Radius $= r$

Height $= h$

Slanted height $= l$

Volume $V = \frac{1}{3}\pi r^2 h$

Surface Area $A = \pi r\left(r + \sqrt{r^2+h^2}\right)$

The contraint is imposed on the Volume and we want to maximize the total surface area. So let's construct a composite function F in terms of V and A. We get

$$F = A + \lambda V$$
$$= \pi r\left(r + \sqrt{r^2+h^2}\right) + \frac{1}{3}\pi\lambda r^2 h$$

Now $F = F(r,h)$

$$\therefore \frac{\partial F}{\partial r} = 0$$

$$\Rightarrow \frac{\partial}{\partial r}\left[\pi r\left(r + \sqrt{r^2+h^2}\right) + \frac{1}{3}\pi\lambda r^2 h\right] = 0$$

$$\Rightarrow 2\pi r + \pi\sqrt{r^2+h^2} - \frac{\pi r^2}{(r^2+h^2)^{3/2}} + \frac{2}{3}\pi\lambda r h = 0$$

$$\therefore 2r + \sqrt{r^2+h^2} - \frac{r^2}{(r^2+h^2)^{3/2}} + \frac{2}{3}\lambda r h = 0$$

And $\frac{\partial F}{\partial h} = 0$

$$\Rightarrow \frac{\partial}{\partial h}\left[\pi r\left(r+\sqrt{r^2+h^2}\right)+\frac{1}{3}\pi\lambda r^2 h\right]=0$$

$$\Rightarrow \quad -\frac{\pi r h}{(r^2+h^2)^{3/2}}+\frac{1}{3}\pi\lambda r^2=0$$

$$\Rightarrow \frac{\pi\lambda r^2}{3}=\frac{\pi r h}{(r^2+h^2)^{3/2}}$$

$$\Rightarrow \frac{\lambda^2 r^2}{9}=\frac{h^2}{(r^2+h^2)^3}$$

$$\Rightarrow (r^2+h^2)^3\lambda^2 r^2=9h^2$$

$$\Rightarrow \lambda^2 r^2\sum_{k=0}^{3}\binom{3}{k} r^{2k}h^{6-2k}=9h^2$$

$$\Rightarrow \lambda^2\sum_{k=0}^{3}\binom{3}{k}\frac{r^{2(k+1)}h^{2(3-k)}}{9}=1$$

$$\therefore \lambda=\left[\sum_{k=0}^{3}\binom{3}{k}\frac{r^{2k+2}h^{6-2k}}{9}\right]^{-1/2}$$ [Somewhat of a relation]

While it is indeed a result. This is not satisfactory enough and the overall relation between r and h might end up requiring a computer to solve.

A bit of a waste of time but fun exercise anyways. Since I couldn't finish the computation.

I am leaving it up to the reader as an excercise. Meanwhile, we can look at another textbook problem from Blundell's Thermal Physics which goes as follows.

Problem:

Maximize the Shannon Entropy $S = -K_B \sum_i P_i \ln P_i$ Subject to constraints $\sum_i P_i = 1$ and $\langle f(x) \rangle = \sum_i P_i f(x_i)$. Show that

$$P_i = \frac{1}{Z(\beta)} e^{-\beta f(x_i)}$$

$$Z(\beta) = \sum_i e^{-\beta f(x_i)}$$

$$\langle f(x) \rangle = -\frac{d}{d\beta} \ln Z(\beta)$$

Sol^n:

Given Shannon Entropy $S = -K_B \sum_i P_i \ln P_i$

Constraints: $\sum_i P_i = 1$ —— (i)

$\langle f(x) \rangle = \sum_i P_i f(x_i)$ —— (ii)

Now the new contructed function For Entropy becomes

$$S' = S + \lambda_1 \sum_i P_i + \lambda_2 \sum_i P_i f(x_i)$$

$$= \sum_i \left[-K_B P_i \ln P_i + \lambda_1 P_i + \lambda_2 P_i f(x_i) \right]$$

$$\therefore \frac{\partial S'}{\partial P_i} = 0$$

$$\Rightarrow \frac{\partial}{\partial P_i} \left[\sum_i \left[-K_B P_i \ln P_i + \lambda_1 P_i + \lambda_2 P_i f(x_i) \right] \right] = 0$$

$$\therefore \sum_i \left[-K_B (\ln P_i + 1) + \lambda_1 + \lambda_2 f(x_i) \right] = 0$$

Since the Summation adds up to zero the invidual terms inside would also add up to zero by cancelling each other out. Which means

$$-K_B (\ln P_i + 1) + \lambda_1 + \lambda_2 f(x_i) = 0$$

$$\Rightarrow \ln P_i = \left(\frac{\lambda_1}{K_B} - 1 \right) + \frac{\lambda_2}{K_B} f(x_i)$$

$$\Rightarrow P_i = \frac{e^{\frac{\lambda_2}{K_B} f(x_i)}}{e^{-(1 - \lambda_1 / K_B)}}$$

$$\therefore \; P_i = \frac{e^{-\beta f(x_i)}}{Z(\beta)} \quad \text{(QED)}$$

PS: Since $K_B, \lambda_1, \lambda_2$ are all constants they can all be absorbed into a specific constant which led to the two new constants here.

One being β which is also known as the inverse temperature given by $\beta = 1/K_B T$.

The other one being Z which is also known as the partition function in Statistical Mechanics. If you don't know much about it yet, don't worry. you will see it later down the line.

The partition function is also incredibly powerful in the sense that it allows us to parameterize all thermodynamic potentials and quantities in terms of itself, the temperature and the volume only. Pretty powerful stuff.

Moving on

We got $P_i = \frac{1}{Z(\beta)} e^{-\beta f(x_i)}$

$$\Rightarrow \sum_i P_i = \sum_i \frac{e^{-\beta f(x_i)}}{Z(\beta)}$$

$$\Rightarrow 1 = \frac{1}{Z(\beta)} \sum_i e^{-\beta f(x_i)}$$

$$\therefore Z(\beta) = \sum_i e^{-\beta f(x_i)}$$ [QED]

And $\langle f(x) \rangle = \sum_i P_i f(x_i)$

$$= \sum_i \frac{e^{-\beta f(x_i)}}{Z(\beta)} f(x_i)$$

$$= \frac{-1}{Z(\beta)} \sum_i e^{-\beta f(x_i)} \left(\frac{\frac{d}{d\beta} e^{-\beta f(x_i)}}{e^{-\beta f(x_i)}} \right)$$

$$= -\frac{1}{Z(\beta)} \sum_i \frac{d}{d\beta} e^{-\beta f(x_i)}$$

$$= -\frac{1}{Z(\beta)} \frac{d}{d\beta} \left(\sum_i e^{-\beta f(x_i)} \right)$$

$$\therefore \langle f(x) \rangle = -\frac{d}{d\beta} \ln Z(\beta)$$ [QED]

With that I am marking the end of this section and continuing onto the following.